

DEVICE INITIALIZATION

On power-up, an initial pause of 200 microseconds is required for the internal substrate generator to establish the correct bias voltage. This must be followed by a minimum of eight active cycles of the row address strobe (clock) to initialize all dynamic nodes within the RAM. During an extended inactive state (greater than 8 milliseconds with the device powered up), a wake up sequence of eight active cycles is necessary to ensure proper operation.

ADDRESSING THE RAM

The ten address pins on the device are time multiplexed at the beginning of a memory cycle by two clocks, row address strobe ($\overline{\text{RAS}}$) and column address strobe ($\overline{\text{CAS}}$), into two separate 10-bit address fields. A total of twenty address bits, ten rows and ten columns, will decode one of the 1,048,576 bit locations in the device. $\overline{\text{RAS}}$ active transition is followed by $\overline{\text{CAS}}$ active transition (active = V_{IL} , t_{RCD} minimum) for all read or write cycles. The delay between $\overline{\text{RAS}}$ and $\overline{\text{CAS}}$ active transitions, referred to as the **multiplex window**, gives a system designer flexibility in setting up the external addresses into the RAM.

The external $\overline{\text{CAS}}$ signal is ignored until an internal $\overline{\text{RAS}}$ signal is available. This "gate" feature on the external $\overline{\text{CAS}}$ clock enables the internal $\overline{\text{CAS}}$ line as soon as the row address hold time (t_{RAH}) specification is met (and defines t_{RCD} minimum). The multiplex window can be used to absorb skew delays in switching the address bus from row to column addresses and in generating the $\overline{\text{CAS}}$ clock.

There are two other variations in addressing the 1M RAM: **$\overline{\text{RAS}}$ -only refresh cycle** and **$\overline{\text{CAS}}$ before $\overline{\text{RAS}}$ refresh cycle**. Both are discussed in separate sections that follow.

READ CYCLE

The DRAM may be read with four different cycles: "normal" random read cycle, page mode read cycle, read-write cycle, and page mode read-write cycle. The normal read cycle is outlined here, while the other cycles are discussed in separate sections.

The normal read cycle begins as described in **ADDRESSING THE RAM**, with $\overline{\text{RAS}}$ and $\overline{\text{CAS}}$ active transitions latching the desired bit location. The write ($\overline{\text{W}}$) input level must be high (V_{IH}), t_{RCS} (minimum) before the $\overline{\text{CAS}}$ active transition, to enable read mode.

Both the $\overline{\text{RAS}}$ and $\overline{\text{CAS}}$ clocks trigger a sequence of events that are controlled by several delayed internal clocks. The internal clocks are linked in such a manner that the read access time of the device is independent of the address multiplex window. However, $\overline{\text{CAS}}$ must be active before or at t_{RCD} maximum to guarantee valid data out (Q) at t_{RAC} (access time from $\overline{\text{RAS}}$ active transition). If the t_{RCD} maximum is exceeded, read access time is determined by the $\overline{\text{CAS}}$ clock active transition (t_{CAC}).

The $\overline{\text{RAS}}$ and $\overline{\text{CAS}}$ clocks must remain active for minimum times of t_{RAS} and t_{CAS} , respectively, to complete the read cycle. $\overline{\text{W}}$ must remain high throughout the cycle, and for time t_{RRH} or t_{RCH} after $\overline{\text{RAS}}$ or $\overline{\text{CAS}}$ inactive transition, respectively, to maintain the data at that bit location. Once

long as the $\overline{\text{CAS}}$ clock is active. When the $\overline{\text{CAS}}$ clock transitions to inactive, the output will switch to High Z.

WRITE CYCLE

The user can write to the DRAM with any of four cycles: early write, late write, page mode early write, and page mode read-write. Early and late write modes are discussed here, while page mode write operations are covered in another section.

A write cycle begins as described in **ADDRESSING THE RAM**. Write mode is enabled by the transition of $\overline{\text{W}}$ to active (V_{IL}). Early and late write modes are distinguished by the active transition of $\overline{\text{W}}$, with respect to $\overline{\text{CAS}}$. Minimum active time t_{RAS} and t_{CAS} , and precharge time t_{PP} apply to write mode, as in the read mode.

An early write cycle is characterized by $\overline{\text{W}}$ active transition at minimum time t_{WCS} before $\overline{\text{CAS}}$ active transition. Data in (D) is referenced to $\overline{\text{CAS}}$ in an early write cycle. $\overline{\text{RAS}}$ and $\overline{\text{CAS}}$ clocks must stay active for t_{RWL} and t_{CWL} , respectively, after the start of the early write operation to complete the cycle.

Q remains High Z throughout an early write cycle because $\overline{\text{W}}$ active transition precedes or coincides with $\overline{\text{CAS}}$ active transition, keeping data-out buffers disabled. This feature can be utilized on systems with a common I/O bus, provided all writes are performed with early write cycles, to prevent bus contention.

A late write cycle occurs when $\overline{\text{W}}$ active transition is made after $\overline{\text{CAS}}$ active transition. $\overline{\text{W}}$ active transition could be delayed for almost 10 microseconds after $\overline{\text{CAS}}$ active transition, $(t_{\text{RCD}} + t_{\text{CWD}} + t_{\text{RWL}} + 2t_{\text{T}}) \leq t_{\text{RAS}}$, if other timing minimums (t_{RCD} , t_{RWL} , and t_{T}) are maintained. D is referenced to $\overline{\text{W}}$ active transition in a late write cycle. Output buffers are enabled by $\overline{\text{CAS}}$ active transition but Q may be indeterminate — see note 16 of AC Operating Conditions table. $\overline{\text{RAS}}$ and $\overline{\text{CAS}}$ must remain active for t_{RWL} and t_{CWL} , respectively, after $\overline{\text{W}}$ active transition to complete the write cycle.

READ-WRITE CYCLE

A read-write cycle performs a read and then a write at the same address, during the same cycle. This cycle is basically a late write cycle, as discussed in the **WRITE CYCLE** section, except $\overline{\text{W}}$ must remain high for t_{CWD} minimum after the $\overline{\text{CAS}}$ active transition, to guarantee valid Q before writing the bit.

PAGE MODE CYCLES

Page mode allows fast successive data operations at all 2048 column locations on a selected row of the 1M dynamic RAM. Read access time in page mode (t_{CAC}) is typically half the regular $\overline{\text{RAS}}$ clock access time, t_{RAC} . Page mode operation consists of keeping $\overline{\text{RAS}}$ active while toggling $\overline{\text{CAS}}$ between V_{IH} and V_{IL} . The row is latched by $\overline{\text{RAS}}$ active transition, while each $\overline{\text{CAS}}$ active transition allows selection of a new column location on the row.

A page mode cycle is initiated by a normal read, write, or read-write cycle, as described in prior sections. Once the timing requirements for the first cycle are met, $\overline{\text{CAS}}$ transitions to inactive for minimum of t_{CP} while $\overline{\text{RAS}}$ remains low